

Bayesian Learning

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Bayesian Learning: Core Idea + Example

Goal: Maintain a distribution over hypotheses and update beliefs using data.

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

Interpretation:

- $P(h)$: Prior belief about model
- $P(D|h)$: Likelihood (fit to data)
- $P(h|D)$: Posterior (updated belief)

Example (Coin Bias):

- h_1 : fair coin ($p = 0.5$), $P(h_1) = 0.9$
- h_2 : biased coin ($p = 0.9$), $P(h_2) = 0.1$
- Data: 9 Heads, 1 Tail

MAP prefers h_2 if likelihood gain outweighs prior.

MAP vs ML + Medical Diagnosis Example

MAP:

$$h_{MAP} = \arg \max_h P(D|h)P(h)$$

ML (uniform prior):

$$h_{ML} = \arg \max_h P(D|h)$$

Medical Test Example:

- $P(C) = 0.008$, $P(+|C) = 0.98$, $P(-|\neg C) = 0.97$
- $P(+|\neg C) = 0.03$

$$P(+) = 0.98(0.008) + 0.03(0.992) = 0.0376$$

$$P(C|+) = \frac{0.00784}{0.0376} \approx 0.21$$

Even accurate tests mislead when disease is rare.

Concept Learning + Version Space Example

Likelihood (consistency model):

$$P(D|h) = \begin{cases} 1 & h \text{ consistent} \\ 0 & \text{otherwise} \end{cases}$$

Posterior with uniform prior:

$$P(h|D) = \frac{1}{|VS|}$$

Example (Study Hours $\rightarrow$ Pass):

- Hypothesis: pass if study $\in [2, 5]$
- Observe: student studied 3 hours and passed

All hypotheses covering “3” survive; others eliminated.

MDL Principle + Decision Tree Example

Occam's Razor: prefer simplest model explaining data.

$$h_{MDL} = \arg \min_h [L(h) + L(D|h)]$$

Example:

- Large tree: 100 rules, 0 errors $\Rightarrow$ cost = 100
- Small tree: 10 rules, 2 errors $\Rightarrow$ cost = 12

MDL selects smaller tree (better generalization).

Link to MAP:

$$\arg \max_h P(D|h)P(h) \equiv \arg \min_h [-\log P(h) - \log P(D|h)]$$

Naive Bayes Classifier + Dataset Example

Assumption:

$$P(a_1, \dots, a_n | y) = \prod_i P(a_i | y)$$

Prediction:

$$y_{NB} = \arg \max_y P(y) \prod_i P(a_i | y)$$

Example (PlayTennis subset):

Outlook	Temp	Humidity	Wind	Play
Sunny	Hot	High	Weak	No
Rain	Mild	High	Weak	Yes
Overcast	Cool	Normal	Strong	Yes

Classify (*Sunny, Cool, High, Strong*) using product of conditional probabilities.

MLE, Cross-Entropy, and Weight Inference

Frequentist ML inference:

$$\hat{w} = \arg \max_w P(y|x, w)$$

Bayesian weight inference:

$$P(w|D) = \frac{P(D|w)P(w)}{P(D)}$$

Example (Linear Regression):

$$y_i = wx_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

MLE $\Rightarrow$ minimize $\sum_i (y_i - wx_i)^2$.

Classification: Negative log-likelihood of Bernoulli gives cross-entropy loss.

Gibbs Algorithm: Approximation to Bayes Optimal

Bayes optimal:

$$\hat{y} = \arg \max_y \sum_h P(y|x, h)P(h|D)$$

Gibbs Approximation:

- 1 Sample $h \sim P(h|D)$
- 2 Predict $\hat{y} = h(x)$

Example:

$$P(h_1|D) = 0.6, P(h_2|D) = 0.3, P(h_3|D) = 0.1$$

Randomly pick a hypothesis and classify.

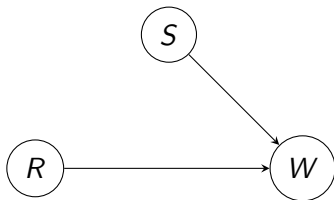
$$E[\text{error}_{Gibbs}] \leq 2E[\text{error}_{Bayes}]$$

Bayesian Belief Networks + Inference Example

Factorization:

$$P(X_1, \dots, X_n) = \prod_i P(X_i | \text{Pa}(X_i))$$

Rain-Sprinkler-WetGrass Network:



$$P(R, S, W) = P(R)P(S)P(W|R, S)$$

Inference Example:

$$P(R = 1 | W = 1) = \frac{P(W = 1 | R = 1)P(R = 1)}{P(W = 1)}$$

Marginalize hidden S to compute $P(W = 1)$.

EM for Gaussian Mixtures + Dataset Example

Hidden variable:

$$z_{ij} = 1 \text{ if } x_i \text{ from Gaussian } j$$

Dataset: $X = \{1, 2, 3, 8, 9, 10\}$ (two clusters)

E-step (soft assignment):

$$\gamma_{ij} = \frac{\pi_j \mathcal{N}(x_i | \mu_j, \sigma^2)}{\sum_k \pi_k \mathcal{N}(x_i | \mu_k, \sigma^2)}$$

M-step (update means):

$$\mu_j = \frac{\sum_i \gamma_{ij} x_i}{\sum_i \gamma_{ij}}$$

Centers iteratively move toward cluster means.

Unified Summary

- Bayesian learning: update beliefs using priors + likelihood
- MAP balances data fit and prior knowledge
- Naive Bayes: simple probabilistic classifier
- MDL implements Occam's Razor
- Gibbs approximates Bayes optimal efficiently
- Bayesian Networks encode conditional dependencies
- EM learns parameters with hidden variables